A Succinct Formalization of the Current Abducer Reasoning Framework

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Abstract

This note formalizes the current implementation state of the Abducer framework as a typed inferential system over mathematical knowledge bases, stage artifacts, and reasoning logs. The formalization is intentionally brief: it records the KB state, the implemented pipeline, the current scoring semantics, and the observability contract, then exemplifies them through one end-to-end theorem-discovery run.

1 Framework Signature

We model the current framework state by the tuple

$$\mathfrak{F} = (\mathcal{K}, \mathcal{A}, \mathcal{M}, \mathcal{T}, \Lambda),$$

where $\mathcal{K}$ is the current problem KB, $\mathcal{A}$ is the structured artifact state, $\mathcal{M}$ is the set of inference modes, $\mathcal{T}$ is the set of deterministic stage operators, and Λ is the reasoning-log trace family.

Definition 1 (KB state)**.** *The current KB state is represented as*

$$\mathcal{K} = (S, C, M, B, R, H, O),$$

where S is the statement set, C the concept set, M the method/proof-regime set, B the barrier set, R the typed relation set, H the proof-hypothesis set, and O the current open obligations.

The derived artifact state is represented as

$$\begin{aligned}\mathcal{A} = (&A^{\text{conn}}, A^{\text{coh}}, A^{\text{ent}}, A^{\text{path}}, A^{\text{hard}}, A^{\text{bridge}},\\ &A^{\text{conj}}, A^{\text{strat}}, A^{\text{sketch}}, A^{\text{eval}}, A^{\text{loop}}, A^{\text{paper}})\end{aligned}$$

2 Current Mode Sets

The current user-facing primary inference modes are

$$\begin{aligned}\mathcal{M}_{\text{primary}} = \{&\texttt{cohere}, \texttt{explain}, \texttt{deduce}, \texttt{abduce},\\ &\texttt{conjecture}, \texttt{proof-sketch}, \texttt{expert-paper}\}\end{aligned}$$

The current prompt-driven auxiliary modes are

$$\mathcal{M}_{\text{prompt}} = \{\texttt{kb-explain}, \texttt{kb-deduce}, \texttt{kb-abduce}, \texttt{kb-induce}\}$$

The current orchestrator intents are

$$\begin{aligned}\mathcal{M}_{\text{orch}} = \{&\texttt{cohere}, \texttt{connect}, \texttt{coherence-eval}, \texttt{interpretation-reports},\\ &\texttt{process-graph}, \texttt{reasoning-dynamics}, \texttt{entrypoints}, \texttt{hardness},\\ &\texttt{bridge-plan}, \texttt{conjecture}, \texttt{strategies}, \texttt{proof-sketch},\\ &\texttt{expert-paper}, \texttt{hypothesis-eval}, \texttt{hypothesis-loop}, \texttt{revision-loop},\\ &\texttt{abductive-loop}, \texttt{pipeline}\}\end{aligned}$$

The composite relation $\mathsf{Cohere} = \mathsf{Coherence} \circ \mathsf{Connect}$ remains the preferred entry operator for relational plus coherentist formation.

3 Pipeline Semantics

The implemented end-to-end pipeline can be written as

$$\begin{aligned}\mathsf{Pipeline} = \mathsf{Paper} &\circ \mathsf{Revise} \circ \mathsf{HypLoop} \circ \mathsf{HypEval}\\ &\circ \mathsf{Sketch} \circ \mathsf{Strategize} \circ \mathsf{Conjecture} \circ \mathsf{Bridge}\\ &\circ \mathsf{Hardness} \circ \mathsf{Path} \circ \mathsf{Entry} \circ \mathsf{Coherence} \circ \mathsf{Connect}\end{aligned}$$

with the revision operator closing the loop by returning a revised KB and revised hypothesis state whenever the hypothesis loop stalls or damages global coherence. The current run state at step t can therefore be written as

$$\mathcal{X}_t = (\mathcal{K}_t, \mathcal{A}_t, \Lambda_t), \qquad \mathcal{X}_{t+1} = \mathsf{Revise}(\mathcal{X}_t) \text{ when revision is triggered.}$$

The core operator family is grouped below by function rather than by stage table, since the framework is best understood through its end-to-end execution rather than a flat catalogue of scripts.

- **Formation layer**: Connect, Coherence, Entry, and Path transform a raw KB into a connected, scored, and ordered inferential frontier.
- **Discovery layer**: Hardness, Bridge, and Conjecture identify the blocking gap and propose the bridge object or abstraction family that would resolve it.
- **Program layer**: Strategize and Sketch turn bridge objects into candidate proof programs with obligations, invariants, and abort conditions.
- **Revision layer**: HypEval, HypLoop, Revise, and Paper evaluate the current bridge set, revise the KB when necessary, and emit the expert-facing synthesis.

4 Worked End-to-End Example

As a concrete witness of the current framework behavior, we use the masked-bridge discovery simulation for the *Markov Kakutani Fixed Point Theorem*. In this run the working KB hides the central bridge concept while the pipeline attempts to recover a proof-relevant route to the target theorem.

$$\mathcal{K}_{\text{ex}}^{\text{work}} = \mathcal{K}_{\text{ex}} \setminus \{\text{explicit bridge-concept annotations}\}, \qquad \mathsf{Pipeline}(\mathcal{K}_{\text{ex}}^{\text{work}}) \rightsquigarrow (\mathcal{A}^\star, H^\star, \Lambda^\star).$$

The selected example uses target statement `st.thm.markov-kakutani`, knockout profile `bridge-concept`, executes 8 iterations, and reaches best replication score 0.909892. If the bridge concept is denoted by $b^\star$, the hidden bridge family in the current run is

$$b^\star \in \{\texttt{cnc.invariant-fixed-point-subset}\},$$

with hidden bridge terms `fixed-point subset`. The recovered discovery chain for the best iteration is:

1. `st.def.compact-convex-set`
2. `st.def.commuting-affine-family`
3. `st.def.fixed-point-subset`
4. `st.lem.cesaro-averages`
5. `st.lem.single-map-fixed-point`
6. `st.prop.fixed-point-subset-structure`
7. `st.lem.commuting-preserves-fixed-point-subset`
8. `st.lem.finite-family-common-fixed-point`
9. `st.thm.markov-kakutani`

The theorem-local conceptual phases recovered by the framework are:

- **Foundational Definitions**: st.def.compact-convex-set (definition): The theorem lives on a nonempty compact convex subset because averaging and compactness both need this ambient geometry.
- **Affine Averaging**: st.lem.cesaro-averages (lemma): Cesaro averages of a single affine orbit stay inside K and provide the averaging mechanism behind single-map fixed-point extraction.
- **Single-Map Fixed-Point Extraction**: st.lem.single-map-fixed-point (lemma): A single continuous affine self-map of a nonempty compact convex set has at least one fixed point.
- **Invariant Fixed-Point Subset**: st.prop.fixed-point-subset-structure (proposition): The [hidden bridge concept] of a single map is itself a nonempty compact convex set.
- **Commuting Invariance**: st.lem.commuting-preserves-fixed-point-subset (lemma): If two affine maps commute, then one preserves the [hidden bridge concept] of the other.
- **Finite-Family Lift**: st.lem.finite-family-common-fixed-point (lemma): One can build a common fixed point for any finite commuting subfamily by iteratively restricting to invariant [hidden bridge concept]s.
- **Common Fixed-Point Theorem**: st.thm.markov-kakutani (theorem): A commuting family of continuous affine self-maps on a nonempty compact convex set has a common fixed point.
- **Examples and Interpretation**: st.ex.interval-commuting-projections (example): On a compact interval, commuting affine self-maps illustrate the theorem in the simplest convex setting.

The example operationally illustrates the full pipeline in four grouped moves:

1. Connect, Coherence, Entry, and Path isolate the highest-leverage statements and expose the missing bridge region rather than treating the theorem as a flat statement-retrieval task.

2. Hardness and Bridge localize the inferential bottleneck as the absent bridge object that connects single-map fixed-point extraction to the finite-family lift.

3. Conjecture, Strategize, and Sketch turn that missing bridge into explicit bridge-object proposals, strategy families, and proof obligations.

4. HypEval, HypLoop, and Revise test whether the proposed bridge improves coherence and reconstructs the target route without reintroducing hidden annotations by hand.

This witness is the intended meaning of the framework signature: the system is not merely scoring isolated hypotheses, but reconstructing a concept-mediated route through a KB whose proof-relevant center has been partially removed.

5 Selection and Scoring Semantics

The framework currently uses stage-specific weighted heuristics rather than a single universal scalar objective. A concise common form is

$$\text{score}(h) = \alpha s(h) + \beta c(h) + \gamma e(h) + \delta \ell(h) + \epsilon n(h) - \rho r(h) - \sigma b(h),$$

where $s(h)$ denotes support, $c(h)$ coherence contribution, $e(h)$ explanatory value, $\ell(h)$ structural leverage, $n(h)$ novelty or fertility, $r(h)$ refutation pressure, and $b(h)$ barrier pressure. The expert-paper stage instantiates an explicit transparent variant of this weighted policy.

6 Observability Contract

Each scripted operator emits a reasoning-log record of the form

$$\lambda_i = (\text{agent}_i, \text{inputs}_i, \text{outputs}_i, \text{summary}_i), \qquad \Lambda = \{\lambda_1, \ldots, \lambda_n\}.$$

This logging contract makes the reasoning process replayable at both the stage level and the run level.

- Connect produces connection-analysis.json/.md, phyp/*.proof-hypotheses.yaml.
- Coherence produces coherence-report.json/.md.
- Entry produces entrypoints.json/.md.
- Path produces entrypoint-pathways.json/.md.
- Hardness produces epistemic-hardness.json/.md.
- Bridge produces bridge-plan.json/.md.
- Conjecture produces concept-gap-abstractions.json/.md.
- Strategize produces proof-strategies.json/.md, novelty prompt.
- Sketch produces proof-sketches.json/.md, paper-style-argument.md.
- HypEval produces hypothesis-evaluation.json/.md, optional phyp/*.hypotheses-evaluated.md, optional updated phyp.

- HypLoop produces hypothesis-refutation-loop.json/.md, phyp/*.hypotheses-reviewed.md, reviewed phyp.
- Revise produces revision-loop.json/.md.
- Paper produces expert-hypothesis-paper.json/.md/.tex.
- Interpret supports the generation of generates human-readable interpretation and discovery-process reports.
- ProcessGraph supports the generation of builds the typed dynamic process graph from run-scoped reasoning logs.
- Dynamics supports the generation of summarizes reasoning dynamics, operator counts, and graph-evolution metrics over a run.

7 Source Basis

This formalization was generated from the current repository documentation sources:

- README.md
- docs/architecture.md
- docs/orchestration.md
- docs/skills.md
- artifacts/markov-kakutani-fixed-point-theorem/discovery-simulation/simulation-report.md
- artifacts/markov-kakutani-fixed-point-theorem/discovery-simulation/iter-008/concept-evoluti
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